

Bluetooth Low Energy (BLE)-Based NEARBY PERSON NOTIFIER

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Imagine the delight of children when they receive a notification of their working mother's arrival while she is still a few steps away from the entry door. Or designing a smart AC that automatically turns on when you are about to arrive home to welcome you with fresh cool air, or a smart garage door that gets notified when you are about to reach and unlocks automatically on your arrival.

There are many such applications where the notifier can be modified and implemented, but here, the notifier is made simple to notify when you are about to reach home to your children. Here, a Bluetooth-based smartwatch is used, and the notifier scans BLE devices. After scanning, it notifies when a particular MAC (media access control) address of the smartwatch comes within the notifier's range.

Bluetooth is ubiquitous today, and various devices and applications on smartphones, smartwatches, tablets, laptops, and iPads opt for this technical advancement. Bluetooth low energy (BLE) is a prevailing technology, and unlike its cousin - the classic Bluetooth technology, it has equally diverse applications. The proposed setup uses BLE to identify the person using the smartphone MAC address and announces their arrival in the setup.

Fig. 1 shows the prototype of the experimental setup and demonstrates the use of BLE. Fig. 2 is the block diagram representation of the entire setup.

BLE operates with the lowest



Fig 1: Authors' prototype of the proposed set up

possible power consumption, is low-cost, low-bandwidth, and simple to use. Such a radio standard was originally initiated by Nokia as Wibree. These design goals eventually led to Bluetooth 4.0 (BLE), a genuine low-power standard that can run for an extended period on a small coin-size cell battery. Bluetooth devices with any specification version prior to 4.0 cannot communicate in any way with a BLE device.

The on-air protocol, the upper protocol layers, and the applica-

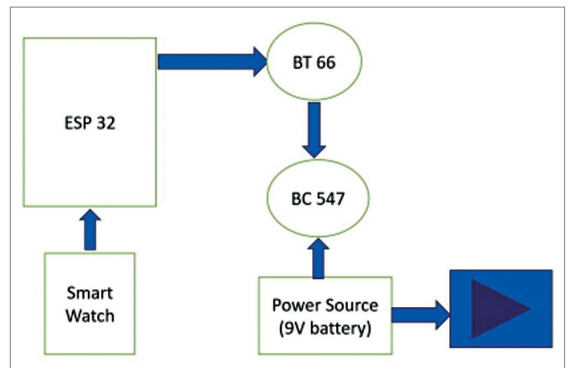


Fig. 2: Block diagram

tions are incompatible between these two technologies. Just like Bluetooth, BLE operates in the 2.4GHz ISM band. Unlike classic Bluetooth, however, BLE remains